# PAVAN KUMAR MATHI

**Software & AI Engineer | Python · LLM Agents · Computer Vision · Full-Stack Web**

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## SUMMARY

Final-year B.Tech ECE student (2027) who builds AI software that runs offline on real devices. Built KAISER, a local AI agent with 41 tools and 492 automated tests; trained a YOLOv8 detector (mAP@50 0.951) for an edge-AI rover; and shipped a live Flask e-commerce site for a local business. Seeking an entry-level software, AI/ML or full-stack developer role.

## TECHNICAL SKILLS

**Languages:** Python, C, JavaScript, SQL, HTML/CSS, Rust (basics)
**AI / ML:** LLM agents, tool calling, RAG, LoRA, Ollama, PyTorch, Transformers, YOLOv8, OpenCV, MediaPipe, Whisper
**Web & Backend:** Flask, SQLAlchemy, PostgreSQL (Neon), REST APIs, Razorpay payments and webhooks, Jinja, Streamlit
**Tools:** Git, GitHub, Vercel, Linux (Debian), Playwright, Google Colab, n8n, Airtable, PuLP

## PROJECTS

**KAISER – Offline Personal AI Agent** | *Python, Ollama, RAG, Whisper, GitHub Actions*

- Voice-activated ("Hey Kaiser") agent that plans a task, asks approval for risky steps, runs it with 41 schema-defined tools and verifies the result, all on local LLMs via Ollama.
- Added background missions (goal, approved plan, step-by-step run, debrief), proactive suggestions, cited web research and PDF question answering with local embeddings.
- Shipped it as a Windows installer with a first-run setup (owner profile, voice enrolment, consent, encrypted API keys); AppContainer sandbox and HMAC audit log; 492 automated tests on CI.

**A.I.R – Artificial Intelligence Rover (Final-Year Project)** | *Python, YOLOv8, Ollama, Arduino UNO Q*

- Trained a YOLOv8n object detector on 592 labelled images (mAP@50 0.951, precision 0.914, recall 0.948) and integrated 8 on-device AI models on a Debian-based Arduino UNO Q.
- Wrote the rover's AI "brain": behaviour modes, offline conversation with Qwen 2.5 1.5B (~2.5 s per reply), Moonshine speech-to-text and Piper text-to-speech.
- Built a 2D simulator where the real AI code drives a virtual MCU; fixed an obstacle-avoidance bug (0 wall collisions over 5 minutes).

**Multimodal Disaster Response Decision Support** | *PyTorch, DistilBERT, ResNet50, PuLP, Streamlit*

- Fine-tuned DistilBERT (15 humanitarian categories) and ResNet50 (damage severity); redesigned the fusion into separate damage and urgency axes after showing naive blending underperformed.
- Allocated rescue teams, ambulances and supplies with fairness-constrained linear programming plus LLM explanations: +24.8% priority-weighted coverage over baseline, under 100 ms per report on CPU.

**MANJULA Ladies Fancy Store – E-commerce Website and Android App (live)** | *Flask, PostgreSQL, Razorpay, PWA*

- Built and deployed a production shop for a local business (ladies-fancy-store.vercel.app): catalogue with WhatsApp enquiries, cart, Razorpay + COD checkout, three delivery modes and an admin panel; installable phone app (PWA) and an Android APK.

**Back Benchers Food Stall – Online Ordering** | *Flask, Razorpay, PostgreSQL*

- Online ordering with Razorpay webhook signature verification and server-side pricing; admin board tracks every order to completion.

**Automation & Vision Tools** | *n8n, Airtable, OpenCV, MediaPipe*

- n8n agent that classifies feedback with an LLM, routes it, logs it to Airtable and emails a reply; OpenCV pipeline with adaptive grayscale and histogram analysis; MediaPipe hand-gesture recognizer for 11 robot commands.

## INTERNSHIPS

**Embedded Systems & IoT Intern – Sreshta Semicon Technologies Pvt. Ltd.** **Jun 2026 – Jul 2026**

- Embedded development and IoT architecture project work, connecting hardware sensor data to software.

**Embedded System Design and IoT Intern (two 30-day internships) – Pantech Prolabs India** **Oct 2026**

**PCB Design & Electronics Intern – Pantech Solutions** **May 2026 – Jun 2026**

## EDUCATION

**B.Tech, Electronics and Communication Engineering – MVGR College of Engineering (A), Vizianagaram** **2023 – 2027**

**Intermediate (MPC) – Sri Chaitanya Junior College, Vizianagaram** **2021 – 2023**

## ACHIEVEMENTS & LEADERSHIP

- Team Lead – ranked 28th of 100+ teams, Autodesk 36-hour hackathon (ANITS, Jan 2026); did the Fusion 360 CAD.
- SPOC Coordinator – organised an electronics workshop at MVGR's Aadhrita tech fest and mentored 50+ participants.

**Languages:** English, Telugu, Hindi